Clean Federated Intrusion Detection Experiment

DNN-EdgeIIoT binary classification | Centralized DNN vs. five-client FedAvg

Objective: compare centralized and federated training after removing likely sources of data leakage.

1. Data preparation

Stage	Result
Original dataset	2,219,201 rows; 63 columns
Original class distribution	1,615,643 normal; 603,558 attack
Conflicting-label rows removed	1,628,811
Same-label duplicates removed	574,563
Clean unique rows before sampling	15,827
Balanced sample used	1,582 rows (10% of clean rows)
Final model features	25 numeric features

2. Final group-disjoint split

Split	Rows	Normal	Attack
Training	1,108	554	554
Validation	236	118	118
Test	238	119	119

Capture groups and exact feature rows do not overlap across splits. Preprocessing was fitted on training rows only.

3. Training configuration

Setting	Value
Architecture	MLP: 25 inputs, hidden layers 128–64–32, 1 output
Centralized training	All training rows together; early stopping on validation F1
Federated training	5 IID clients; 2 local epochs; 10 FedAvg rounds
Federated selection	Best validation-F1 checkpoint: round 3
Learning rate / batch size	0.001 / 256

4. Federated client distribution

Client	Rows	Normal	Attack
Factory A	222	111	111
Factory B	222	111	111
Factory C	222	111	111
Factory D	222	111	111
Factory E	220	110	110

Experiment Results

Both approaches were evaluated on the same untouched, balanced test set of 238 rows.

5. Test-set comparison

Metric	Centralized	Federated
Accuracy	97.06%	97.06%
Precision	99.12%	99.12%
Recall	94.96%	94.96%
F1 score	97.00%	97.00%
ROC-AUC	99.17%	99.24%
PR-AUC	99.35%	99.43%
False-positive rate	0.84%	0.84%
False-negative rate	5.04%	5.04%

6. Test confusion-matrix results

Outcome	Centralized	Federated
Normal correctly classified	118	118
False alarms	1	1
Attacks missed	6	6
Attacks correctly detected	113	113

7. Main findings

Finding	Summary
Accuracy target	Reached: both models exceeded the 96% target
Federated outcome	FedAvg matched the centralized model's accuracy and F1
Error profile	1 false alarm and 6 missed attacks for both models
Interpretation	IID, balanced clients and full participation made the comparison favourable

Conclusion and limitation

The five-client FedAvg simulation matched centralized training under clean, balanced IID conditions.

This was a single-machine educational simulation; preprocessing was centralized.

Privacy and secure-aggregation mechanisms were not implemented.